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Electrode Flexibility Enhances Electrolyte Dynamics during Supercapacitor Charging

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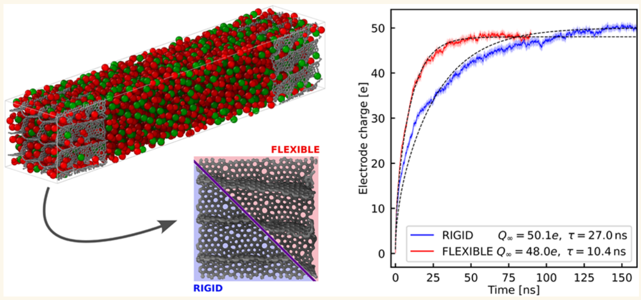

ABSTRACT: **Supercapacitors are energy storage devices with high power density and long cycle life. Combined with spectroscopy and electrochemistry, molecular simulations and theory have allowed to characterize their charging mechanisms, and we now have an excellent understanding of the effect of properties such as nanopore size or structural disorder on the supercapacitor performances. However, the influence of electrode flexibility remains to be addressed as state-of-the-art models focused on the polarization effects of the electrode, but enforced its structural rigidity, as an approximation. Here we overcome this limitation by integrating a constant-potential molecular dynamics scheme with a state-of-the-art machine-learning potential for carbon, while controlling the applied potential. Using nanoporous sp^2/sp^3 carbon electrodes filled with an ionic liquid electrolyte, we compare the behavior of the rigidified and flexible frameworks; the latter allowing for local atomic relaxation, breathing modes, etc. We demonstrate that flexibility significantly enhances in-pore ionic diffusivity, thus shortening the characteristic charging time by a factor of 3 relative to the rigid analogue, while the specific capacitance remains in the experimental range (≈140 $F \cdot g^{-1}$) for both cases. More specifically, our analyses demonstrate that flexibility accelerates co-ion expulsion, mitigates pore overcrowding, and promotes a homogeneous induced-charge profile that penetrates deeply from the very start of the applied voltage onset, explaining the improved kinetics.**

KEYWORDS: *machine learning, flexible, electrode, molecular dynamics, constant potential, supercapacitors*

INTRODUCTION

In supercapacitors, the electricity is stored via an ionic adsorption mechanism that is highly reversible and fast. Contrarily to batteries, there are no redox reactions and everything occurs at the surface of the electrode, making supercapacitors the devices of choice for high-power applications. Spurred by the discovery of the importance of nanopores in the charging mechanism,[1] the fundamental research devoted to understanding the underlying mechanism has hugely increased over the past two decades. In order to complement electrochemical measurements, spectroscopy techniques were adapted to probe *in situ* the behavior of the ions within nanopores.[2] Simulation methods, such as molecular dynamics, also had to be augmented to account for the polarization of the electrodes upon application of an electrical potential between them.[3] Combining these approaches allowed for a detailed exploration of charging mechanisms, leading to a good understanding of the structure−property relationship in carbon-based supercapacitors[4] as well as in more complex nanoporous materials such as metal organic frameworks[5,6] or two-dimensional (2D) materials.[7,8]

The main remaining challenge is to understand qualitatively the dynamic processes occurring upon supercapacitor charging. Dynamic effects were initially less explored as the typical characteristic times for charging a structure on the nanoscale are rather long, requiring lengthy and expensive computer simulations to be performed.[9,10] These simulations showed that complex collective phenomena are at play. For example, molecular dynamics calculations displayed an overcrowding mechanism within the pores, that leads to local slow-down of the diffusion, calling for the development of new charging protocols to enhance the charging rate.[11] However, it is

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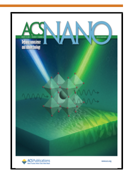

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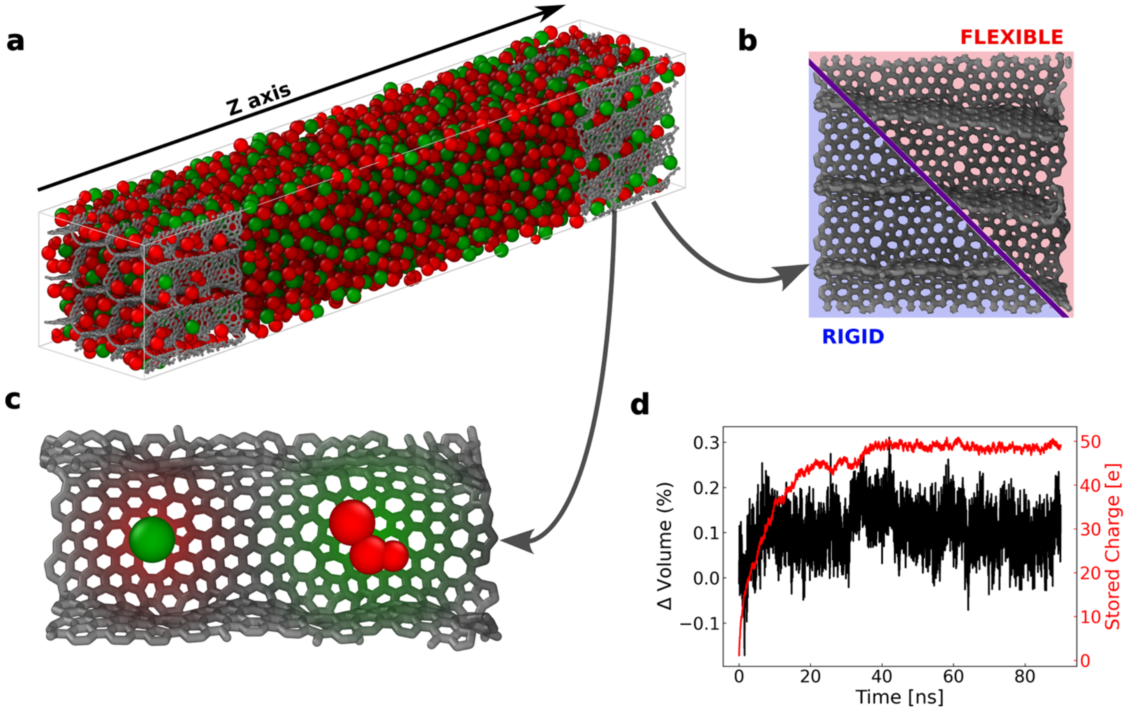

Figure 1. Flexible and polarizable supercapacitor model. (a) A snapshot of the simulated system showing the graphitic nanoporous electrodes (gray) and the ionic liquid electrolyte (red and green spheres). (b) Slice along the *xz* plane of the rigid and flexible electrode structures. (c) Snapshot of the induced charge distribution on electrode atoms under potential difference near an anion (green) and a cation (red). (d) Induced charge on the cathode (red) and relative volume variations of the simulated system (black) during the charge for a trajectory.

standard in constant-potential MD studies to freeze the carbon framework because accurate, affordable forces were unavailable until recently: ab initio calculations were too costly for the large cells needed, while classical force-fields could not reproduce the elastic response of disordered carbons. This limitation is concerning since experiments have shown that electrosorption of the ions may impact the local nanostructure of the electrodes, thus opening the possibility for additional storage mechanisms.[12,13] Earlier studies have also demonstrated that the mechanics of porous carbons can influence how many ions a supercapacitor stores. *In situ* dilatometry, for instance, revealed that charging a carbide-derived carbon film in an ionic liquid produces anisometric swelling of up to 2%.[14,15] Theoretical models[16] for battery systems suggest that electrode flexibility affects pore dynamics, influencing system capacity and ionic mobility.

A recent study tried to address this limitation through the use of a reactive force field for the carbon electrodes.[17] Nevertheless, the obtained capacitances for the flexible models were 1 order of magnitude lower than the experimental values, which may be due to the use of a highly disordered kerogen-like structure for the carbon electrode, while supercapacitors with good performances generally involve disordered carbons with small graphene-like domains.[18] The advent of machine learning potentials (MLPs) opens new perspective for the simulation of supercapacitors. In particular, in the case of carbon, several MLPs were developed with the ability to match density functional theory (DFT) accuracy for structure, energetics, and elasticity while operating orders of magnitude faster, making fully flexible electrodes practical in large-scale supercapacitor simulations.[19−22] Exploiting these potentials, we allow every electrode atom to move under a difference of potential and thus capture the electro-mechanical coupling that rigid-wall models miss.

RESULTS AND DISCUSSION

Supercapacitor Model. The supercapacitor system we simulate is composed of nanoporous carbon electrodes with tubular pores oriented toward the *z*-axis and a coarse-grained model of the 1-butyl-3-methylimidazolium hexafluorophosphate (BMIM PF_6) ionic liquid (Figure 1(a)). This nanomaterial structure was chosen because it displays a single type of nanopore, which simplifies the rationalization of the results, while being largely made of graphene-like domains. Simulations were performed using the constant potential method.[23,24] In this approach, each carbon atom carries a Gaussian charge distribution, which intensity fluctuates depending on the surrounding electrolyte atoms in order to maintain a fixed voltage between the two electrodes.

Interactions between carbon atoms in the electrodes are described using a parametrization of the Atomic Cluster Expansion MLP[25] for carbon allotropes,[22] which was demonstrated to accurately capture various carbon structures and their dynamic properties − note that no additional electrostatic interactions are included between pairs of carbon atoms by the MLP. Two models were compared: the rigid model, serving as the state-of-the-art reference, in which the electrode atoms remain fixed and trajectories are sampled within the *nVT* ensemble, and the flexible model, in which the movement of electrode atoms is explicitly computed. This has two consequences: first, the atoms are allowed to relax, which leads to local changes in the porous structure of the electrodes; Second, it enables the use of a barostat to sample trajectories from the *nPT* ensemble, allowing for collective vibrational

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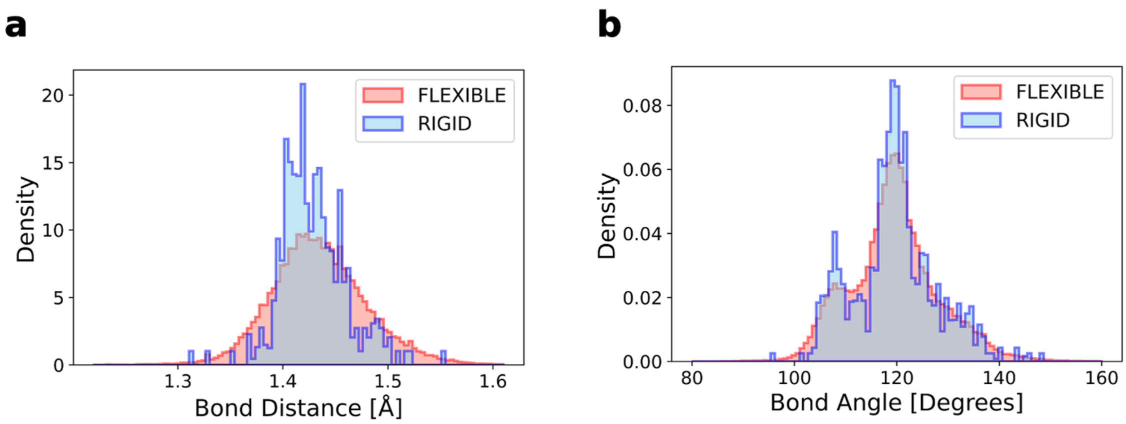

Figure 2. Density histograms of the bond distances (a) and bond angle (b) of the carbon atoms composing the electrodes for both rigid and flexible models.

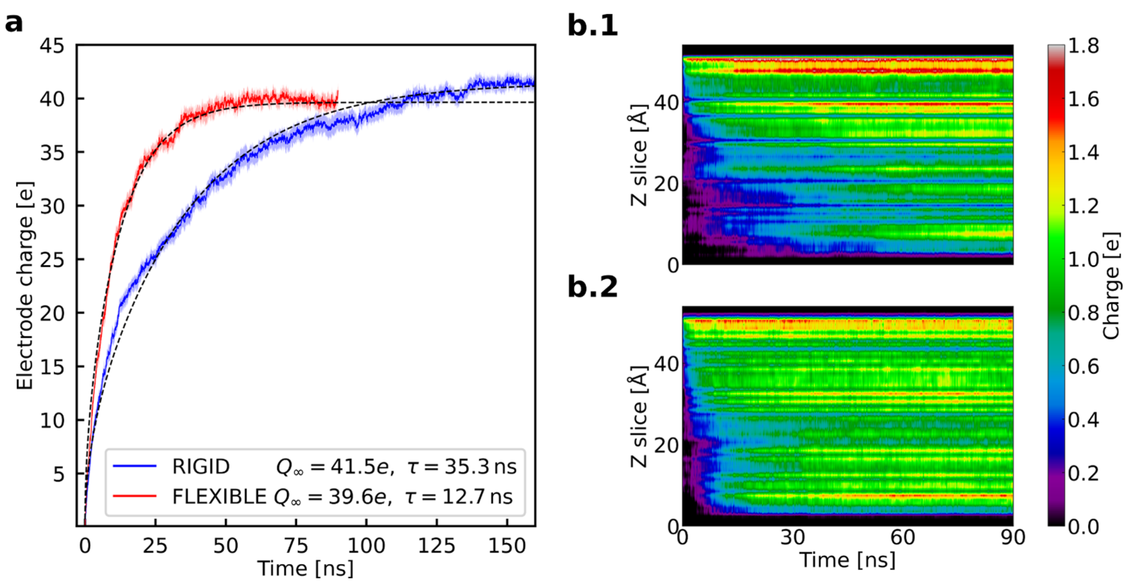

Figure 3. Comparion of the charging dynamics in the rigid and flexible electrode models. (a) Total induced charge on the electrodes versus time for the rigid and flexible electrode models under an applied potential difference of 1 V. Solid lines represent simulation results, while dotted lines show the fitted behavior using the transmission line model. (b) Distribution of the charge induced per layer in the cathode over time for the rigid (b.1) and flexible (b.2) models. In real systems, the current collector is located at z = 0.

modes — such as breathing— of the electrode during electrosorption of the ions. A constant potential difference of 1 V was imposed between the electrodes.

As shown in Figure 1(b), the global structure of the electrode, including the location of defects in the graphitic surfaces and the joints between tubular pores, is preserved in the flexible model. In addition, expansion or distortion of pores are allowed, resulting in broader, more realistic bond angle and distance distributions of the structure (Figure 2(a,b)), in the sense that temperature effect can now be captured. The stability of structural features such as the angle and distance distributions in the electrodes reflects the reliability of the MLP for this system.

Charging Dynamics. The main difference between the two models lies in the charging dynamics of the electrodes, as shown in Figure 3(a). The time evolution of the total electrode charge with time was fitted using a transmission line model (TLM) with an infinite RC ladder as equivalent circuit to model the electrode−electrolyte interface in the pores,[26] resulting in the following equation

$$Q(t) = Q_\infty\left(1 - \frac{2}{\pi^2}\sum_{n=0}^{\infty}\frac{\exp\left[-\pi^2\left(n+\frac{1}{2}\right)^2\left(\frac{2l}{L}\right)^2\frac{t}{\tau_c}\right]}{\left(n+\frac{1}{2}\right)^2}\right) \tag{1}$$

in which $1/l$ is the mean surface area per volume of the pores in $Å^{-1}$ and L is the pore length in Å. When the induced charges have reached equilibrium, the fit yields two quantities, the total induced charge at equilibrium Q_∞ and the characteristic time (τ_c), from which we can retrieve the characteristic TLM charging time $\tau = L^2/(\pi l)^2\tau_c$. The flexible model is characterized by a charging time close to three times shorter than the rigid one (Figure 3a), indicating largely enhanced power density. On the other hand, the total electrode charges differ only slightly, the flexible model one being 5% lower than the rigid one. From this value it is possible to estimate the electrode capacitance $C = Q_\infty/\Delta\Psi$, where $\Delta\Psi$ is the potential difference between the bulk liquid and the electrode. If we assume that the applied voltage shares evenly between the two

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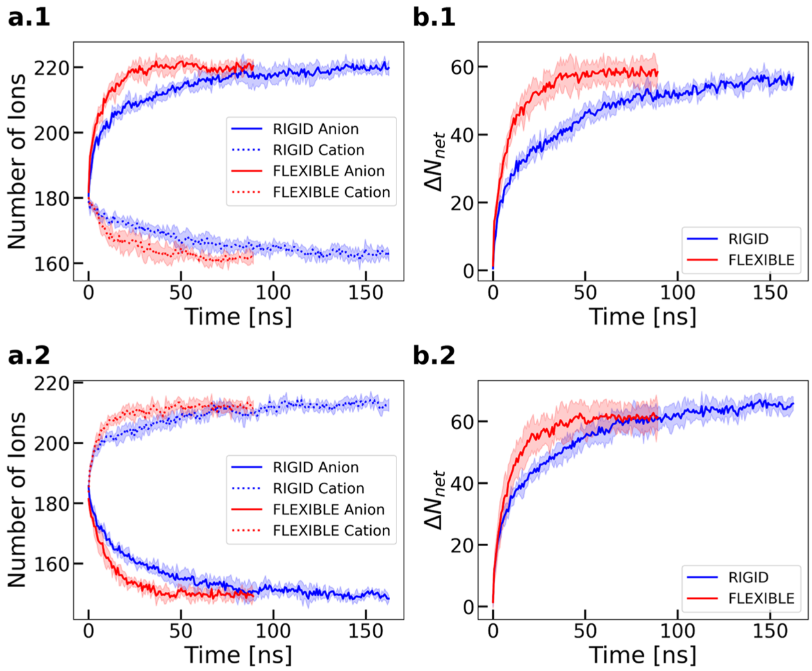

Figure 4. Ion dynamics within the electrodes. Time evolution of the: (a) number of counterions (solid lines) and co-ions (dashed lines) in the cathode (a.1) and anode (a.2) over time; (b.1, b.2) net ion exchange parameter (ΔN_{net}).

electrodes (which is a reasonable approximation in this system[3]), this yields specific capacitances $C_m = C/m$ (m being the mass of the electrode) of 146 and 139 F g^{-1} for the rigid and flexible models respectively, which agrees with experimental results with such pore sizes.[1] This is at odd with the recent work on the adsorption of aqueous electrolytes within flexible kerogen-like materials, in which a much smaller value of 16 F g^{-1} was obtained.[17] Although this was attributed by the authors to the concentration of the electrolyte, such effects were not reported in previous works focused on capacitive desalination[27] and the discrepancy is more likely to be due to the structure of the carbon itself.

The effects of volume variation and structural relaxation were first examined by simulating the charging process with rigid, relaxed electrode structures (Supporting Figure 2). These tests showed that a rigidified electrode set to the final volume of the flexible model exhibits charging dynamics nearly identical to those of the original rigid model. In addition, no direct correlation was observed between the simulated box volume and the induced charge in the flexible model (Figure 1d). Taken together, these evidence indicate that the contrasting behavior of the rigid and flexible models is driven by electrode flexibility rather than by volume variations. The spatial distribution of induced charges within the cathode reveals distinct behaviors as shown in Figure 3(b). In the rigid model, the outer electrode layers rapidly acquire most of the initial charge, followed by a more gradual (and slow) penetration of charge into deeper layers over time, in agreement with previous work.[10] This process leads to a pronounced layering effect within the electrode, accompanied by oscillations in ionic charge density that arise from the preferential adsorption sites for ions. In stark contrast, the flexible model immediately produces a much more homogeneous charge distribution across the electrode layers from the outset. The differences in charge amplitude between individual layers are largely reduced, and charge penetrates the deeper regions of the electrode faster. This homogenization is consistent with the dynamic motion of electrode atoms, which helps facilitate more uniform ionic interactions throughout pores. This shows that the accelerated charging we observe is not a finite-size effect due to the nanometric size of the electrode, but that we can expect it to persist over much larger depth within larger electrodes.

Electrolyte Dynamics. We now analyze more precisely the dynamic mechanisms occurring during electrosorption. Figure 4 illustrates the temporal evolution of ion populations inside the nanoporous electrodes, and highlights the role of electrode flexibility in enhancing ion mobility and charge equilibration.

In Figure 4(a.1),(a.2), the numbers of anions and cations within each electrode are tracked as a function of time. The rapid rise in counterion adsorption in the flexible model contrasts with the more gradual increase observed in the rigid model. This is made possible by the fast co-ion desorption (Figure 4(a.1)), which has already been identified as the rate-limiting step in previous studies.[28] The desorption of co-ions out of the electrode is accelerated when flexibility is included, leading to a swifter co-ion depletion compared to the rigid model. Another competing effect observed here is related to the geometry of the ions. Indeed, the bulkier and anisotropic cations experience stronger crowding effects compared to the spherical anions, making the co-ion desorption even slower in the cathode and balancing the sorption rates in the anode. The effect of flexibility is therefore exacerbated in the cathode, as shown by the net ion exchange parameter, $\Delta N_{net} = \Delta N_{ads}^{counterion} - \Delta N_{ads}^{co\text{-}ion}$, depicted in panels (b.1) and (b.2). $\Delta N_{ads}^{counterion}$ (respectively, $\Delta N_{ads}^{co\text{-}ion}$) is defined as the variation of adsorbed counterions (respectively, co-ions) between time t and the initial configuration. In both electrodes, ΔN_{net} increases more rapidly under the flexible conditions, suggesting that electrode motion facilitates ionic migration pathways. This effect coincides with an enhanced overall ionic conductivity and more efficient charge balancing.

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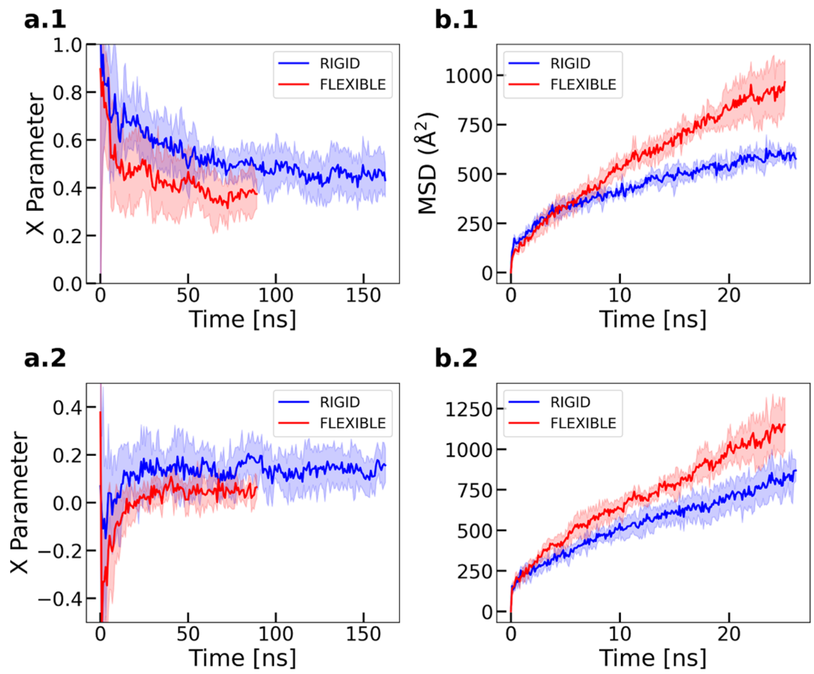

Figure 5. Ion exchange balance parameter (X) in the cathode (a.1) and in the anode (a.2); (b) mean-squared displacements of anions in the cathode (b.1), and cations in the anode (b.2).

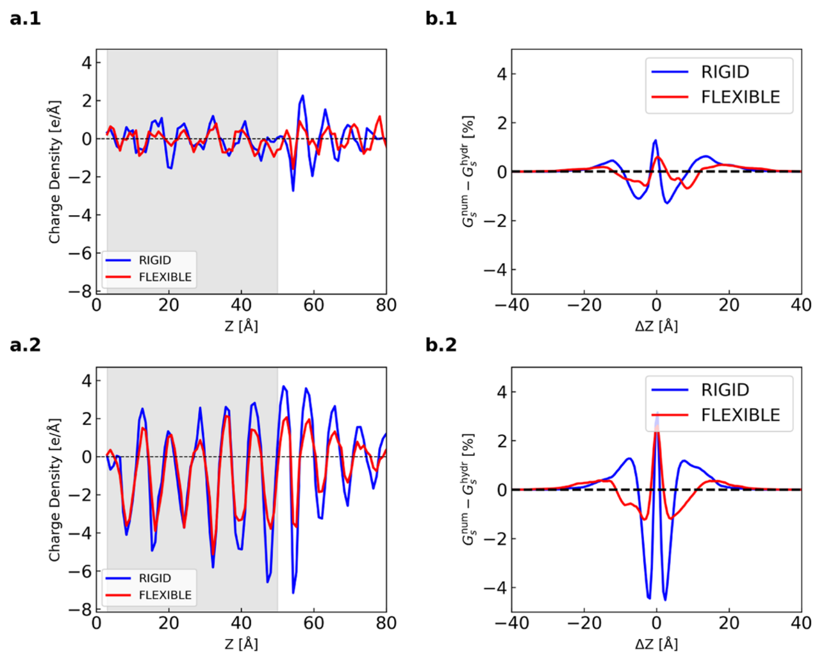

Figure 6. Structuration of in-pore ionic densities and dynamics. (a) ionic charge density profiles, projected along the z axis, and averaged on the initial (a.1) and final (a.2) 2 ns of simulation. (b) Difference between the Van Hove function and the corresponding hydrodynamic limit of the co-ions in the cathode (shaded region) on the z axis for Δt = 20 ns, computed at the beginning of the charging process (b.1) and at its end, when the system is completely charged (b.2).

$$X = \frac{\Delta N_{\text{ads}}^{\text{counter-ion}} + \Delta N_{\text{ads}}^{\text{co-ion}}}{\Delta N_{\text{ads}}^{\text{counter-ion}} - \Delta N_{\text{ads}}^{\text{co-ion}}} \tag{2}$$

To quantify the balance between counterion adsorption and co-ion desorption events, the charging mechanism parameter is introduced[29] (see also Supporting Figure 1). The X parameter remains positive in the cathode throughout charging (Figure 5a.1), confirming the counterion adsorption as the dominant process. In the anode, X is briefly negative during the first few nanoseconds, indicating a rapid co-ion desorption, before turning positive as adsorption eventually dominates. For both electrodes, X stabilizes at a lower plateau in the flexible model, whereas the rigid model shows larger positive excursions. Likewise, the mean-squared displacements of anions (b.1) and cations (b.2) rise more steeply in the flexible model, demonstrating higher ion mobility.

Taken together, these observations reveal that electrode flexibility accelerates counterion adsorption, facilitates co-ion desorption, and yields a more balanced ion-exchange mechanism. This behavior explains the superior charging performance observed for the flexible electrode supercapacitor.

Heterogeneous Dynamics. In order to further characterize the overcrowding mechanism within the cathode, the one-dimensional ionic charge density profile along the z axis, averaged over the initial and the final 2 ns of the simulation, are respectively displayed in Figure 6(a.1),(a.2). These charge densities show alternating anion and cation layers in the cathode. The rigid model (blue) exhibits higher charge separation, as indicated by the larger amplitude of charge density per slice compared to the flexible model (red). Gray areas correspond to electrode regions. The rigid model exhibits a more pronounced layering of ions with larger peaks, reflecting stronger charge separation effects. In contrast, the flexible model shows lower-amplitude oscillations, which indicates a weaker structuration of the adsorption sites, and reduced overall charge separation. These differences become especially apparent in the electrode regions (shaded in gray), where structural flexibility, under the effect of the temperature, enables the carbon atoms to adjust and promote more uniform ion distributions.

In statistical physics, heterogeneous dynamics is often analyzed using the self-part of the Van Hove function ($G_s(\Delta r, \Delta t)$), which measures the probability of finding a particle at a given displacement Δr away from its initial position after a time interval Δt. Under purely hydrodynamical (i.e., ideal or "free") diffusion, this function follows a Gaussian distribution whose width grows in time according to the diffusion coefficient.[30] As a consequence, comparing G_s from simulations to the corresponding hydrodynamical limit reveals how intermolecular interactions and structural constraints modify the diffusion process. Here, since the charging occurs along the z axis, we limit our analysis to this dimension and calculate $G_s(\Delta z, \Delta t)$. In Figure 6(b.1),(b.2), the difference between the simulated G_s^{num} and the hydrodynamical limit G_s^{hydr} shows that the rigid model exhibits significantly larger deviations, as a result of its more pronounced layered ionic motion. On the other hand, the flexible model exhibits smaller discrepancies, suggesting that ions in this scenario diffuse more similarly to the ideal, free-diffusion behavior, while being less constrained by the electrode structure.

To illustrate the differences in diffusion behaviors, Figure 7(a),(b) display the trajectories of the 20% deepest co-ions in the cathode, with colors corresponding to their initial z-position. The trajectories reach up to ≈40 Å, i.e., near the exit of the pore. In the flexible electrode, these co-ions diffuse easily through the porous network, indicating an environment that favors rapid ion movement. Such enhanced ionic conductivity when accounting for flexibility, which matches the predictions of our transmission line model, is directly connected to decreased preferential adsorption sites, a reduction in charge separation that facilitates ion transport, and the capacity for structural rearrangements that induce faster ion mobility. Collectively, these results underline the structural and dynamical advantages provided by electrode flexibility, ultimately contributing to improved supercapacitor performance. Beyond this result, the differences observed between the rigid and flexible simulations demonstrate that the 'frozen electrode' approximation can limit our understanding of the qualitative mechanisms controlling the charging phenomena.

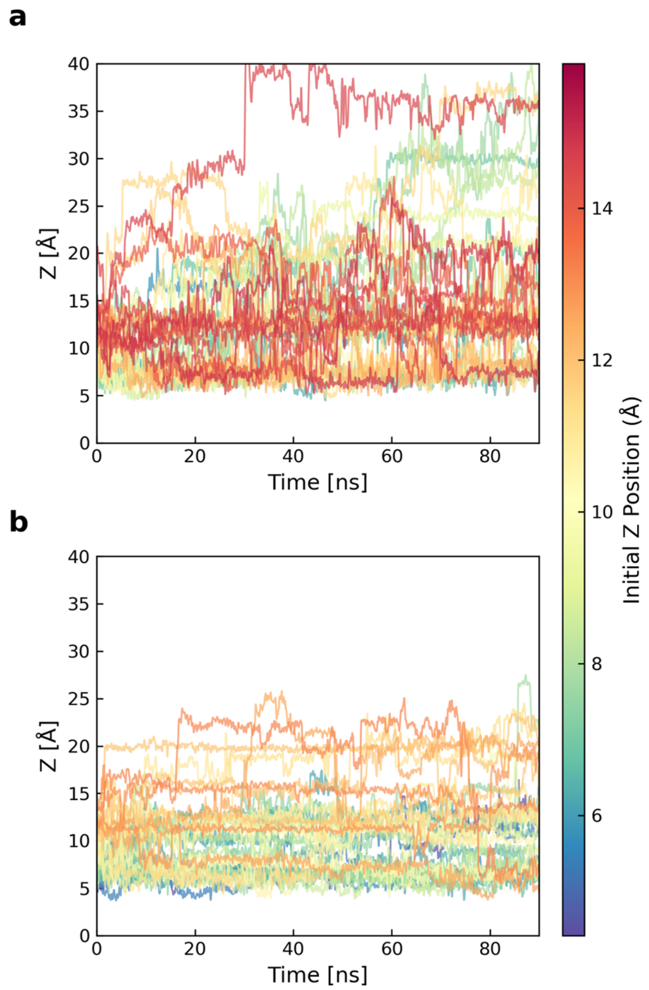

Figure 7. Trajectories of the 20% deepest co-ions projected over the z axis, for the flexible (a) and rigid (b) models, inside the cathode (5 Å < z < 50 Å).

CONCLUSIONS

In conclusion, our findings demonstrate that electrode flexibility significantly impacts the estimation of the electrode charging times, and should therefore be mandatarily be taken into account when studying the power density of supercapacitors using molecular simulations. The flexible model allows structural adjustments that promote homogeneous charge distribution and faster ion rearrangement upon application of a potential. The enhanced dynamics is not only reflected in the charging times, but also in more qualitative traits such as sorption mechanisms, crowding of the pores, and the individual in-pore diffusivities. We would like to stress, however, that the monodisperse ordered pore model used here was deliberately kept simple. This idealized geometry allowed us to isolate the electro-mechanical coupling mechanisms and interpret the results unambiguously, at the cost of omitting the structural disorder that characterizes real nanoporous carbons. Adapting our methodology to the more complex, flexible, and size-disordered electrodes is an interesting direction for future work; the framework we established in the present study provides the technical foundation for such future investigations of strain and pressure-controlled charging in realistic materials.

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This work has implications for the *in silico* design of next-generation supercapacitors, which aims at engineering the electrode structure and composition for optimal performance. The recent development of highly accurate MLP models of materials that display promising electrochemical performances for batteries applications such as metal−organic frameworks,[31] metals or layered nanomaterials,[32] opens the way for systematic studies on the effect of electrode materials flexibility on the properties of supercapacitors. Going beyond, they should enable the simulation of realistically complex electrolytes such as hydrogels,[33,34] for which the mechanical properties can also play an important role on the ion adsorption mechanisms and dynamics.

METHODS

All simulations were performed using the LAMMPS[35] molecular dynamics package, in particular with the ELECTRODE module[36] for constant potential simulations. The supercapacitor model was retrieved from a previous work;[3] the electrodes have been generated using simulated annealing with a parametrization of the GAP MLP for carbonaceous systems.[37]

Two electrode models were employed to investigate the effect of structural flexibility on supercapacitor performance. For statistical significance, five independent molecular dynamics simulations were performed for each model. Physical observables shown in the main text were systematically averaged over those five trajectories, and errors reported correspond to the estimated standard deviation of the observables. Simulations were respectively run for 163 and 90 ns for the rigid and flexible models, which corresponds to convergence in the induced charge on the electrodes. All simulations were performed at 400 K. For the rigid model, we performed simulations in the *nVT* ensemble using a chain of three Nosé-Hoover thermostats. For the flexible model, we sampled the *nPT* ensemble at atmospheric pressure, using a Berendsen barostat and the same chain of three Nosé-Hoover thermostats. One single atom of each electrode has been fixed in the flexible model to prevent the electrode structures from drifting in the xy plane. All the simulations were run using a time step of 2 fs. The simulation box is periodic in the x and y directions, but not in the z direction; to confine the particles within the box, a Lennard-Jones wall with parameters corresponding to carbon was placed on both ends along the z axis.

Charging curves were obtained by measuring the induced charge on the cathode over time under a 1 V constant potential difference. The charge shown in Figure 3a has been computed only on the first 45 Å of the electrode, neglecting the interfacial layer. This procedure does not change the interpretations and relations between quantities obtained in the two models as shown in the Figure 2 of Supplementary but enhances the quality of the fit for the rigid model which interfacial layer has a specific behavior. The characteristic charging time was extracted using a transmission line model fit. Ion dynamics were analyzed by tracking adsorption and desorption events, and mean-squared displacement (MSD) and Van Hove functions calculations were performed to assess ionic mobility.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.5c07490.

> Additional analyses including a schematic description of the different charging mechanisms characterized by the parameter X, charging curves comparing rigid, rigidified, and flexible electrodes, and methodological details on the treatment of electrostatics and machine-learning potentials. These results support the main conclusions by clarifying the role of structural dynamics and force contributions during electrode charging (PDF)

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Author Contributions

Z.W., A.F.L., M.S. and A.M.S. designed the research. Z.W. carried out simulations and analysis of the data. All authors discussed the scientific results and edited the manuscript.

Notes

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The authors declare no competing financial interest.

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